

Understanding the Generative Echo

Chamber effect ~~cognitive susceptibility~~ during voice-based conversational search

 The objective of this study is to understand how user's cognitive susceptibility changes with voice modality of conversational search ~~Large Voice Models (LVMs)~~ and what are the reasons for this change. A secondary objective is to study the effectiveness of current solutions designed for text based conversational search and design solutions to address the gaps found.

~~The objective of this study is to understand the distinctive features of the over reliance problem with Large Voice Models (LVMs), identify the effectiveness of current solutions in voice modality and finally to propose new solutions to handle the new voice modality of LLMs.~~

 Users of AI systems often face the challenge of skewed trust and over reliance on the system which lead to increased cognitive susceptibility. This can lead to many downstream problems from propagation of misinformation to even harm to the user due to harmful advice^[1,13]. The question then becomes how can LLM verbalize its uncertainty and more importantly probe the user to explore the topic further. However, as seen with text based LLMs this is often difficult due to a plethora of reasons including but not exhaustive of the persuasive power of LLM generated text, increased cost to the user to probe and verify the information and LLMs being bad at verbalizing their uncertainty.

💡 Past literature has explored the problem of over-reliance in LLMs ^[2-10] extensively and has found the problem challenging to solve. Although not a complete solution, past work has hinted that verbalizing uncertainty through natural language showed decreased user trust in the system ^[2,11,12]. There are also studies that focus on providing citations and references to curb over-reliance ^[16-18]. However, as highlighted by other papers just having citations and quotes can have the reverse effect too and increase over reliance.

Looking at cognitive susceptibility, literature has shown phenomena such as Generative Echo Chambers ^[20,21] where users have shown to be influenced by confirmatory AI leading to increased polarization. Some HCI studies have tried to tackle the issue of selective exposure and diverse information seeking through UX redesign but none have paid dividends ^[22].

Voice based interactions add new dimensions to interactions with users along with a decreased cognitive load ^[23]. The decreased cognitive load for the user and added expressiveness to LLMs coupled with the environment of the interaction with voice modality can lead to significant increase in cognitive susceptibility which we want to measure.

~~Voice modality adds a new dimension in the interaction with users. Voice gives the LLM more expressiveness and for humans voice gives us an understanding of the person's emotions, personality, confidence, etc. We want to compare the problem of over-reliance in text modality to that of voice modality and analyze the effectiveness of current solutions.~~

The environment of the interaction in voice modality can be either passive (limited cognitive resources) or active. There is an environment shift in users' interaction with text based assistants vs voice based assistants. Usually, the use of voice based assistants is accompanied by another overlapping activity because of which the user is not able to use a text based medium. We want to study how users interact and engage with agents in such environments and highlight the nuances of such interactions when looking at cognitive susceptibility.

★ The research questions we want to answer are as follows:

- RQ1: How does a user's cognitive susceptibility change when interacting with conversational search in voice vs text modality ~~LVMs vs LLMs?~~
- RQ2: Does the familiarity of the topic change the effect?
- RQ3: Does being in an environment where there are limited cognitive resources available exacerbate the cognitive susceptibility?
- RQ4: What solutions currently in place are deployable in a voice modality and what is their effect?
- RQ5: What are potential solutions forward?
- ~~● RQ1 1: How differently over-reliance and trust manifest in voice based LLM interactions compared to text based LLM interactions.~~
- ~~● RQ1 2: How does the environment in which voice based interactions take place change this effect?~~
- ~~● RQ2: How effective are current solutions for over-reliance in voice modality?~~
- ~~● RQ3 1: What are the unique ways in which LLMs can verbalize their uncertainty in the voice based interactions?~~
- ~~● RQ3 2: How effective are these approaches?~~

Q: Do we allow people to choose the topic? No

Q: Will people ask questions

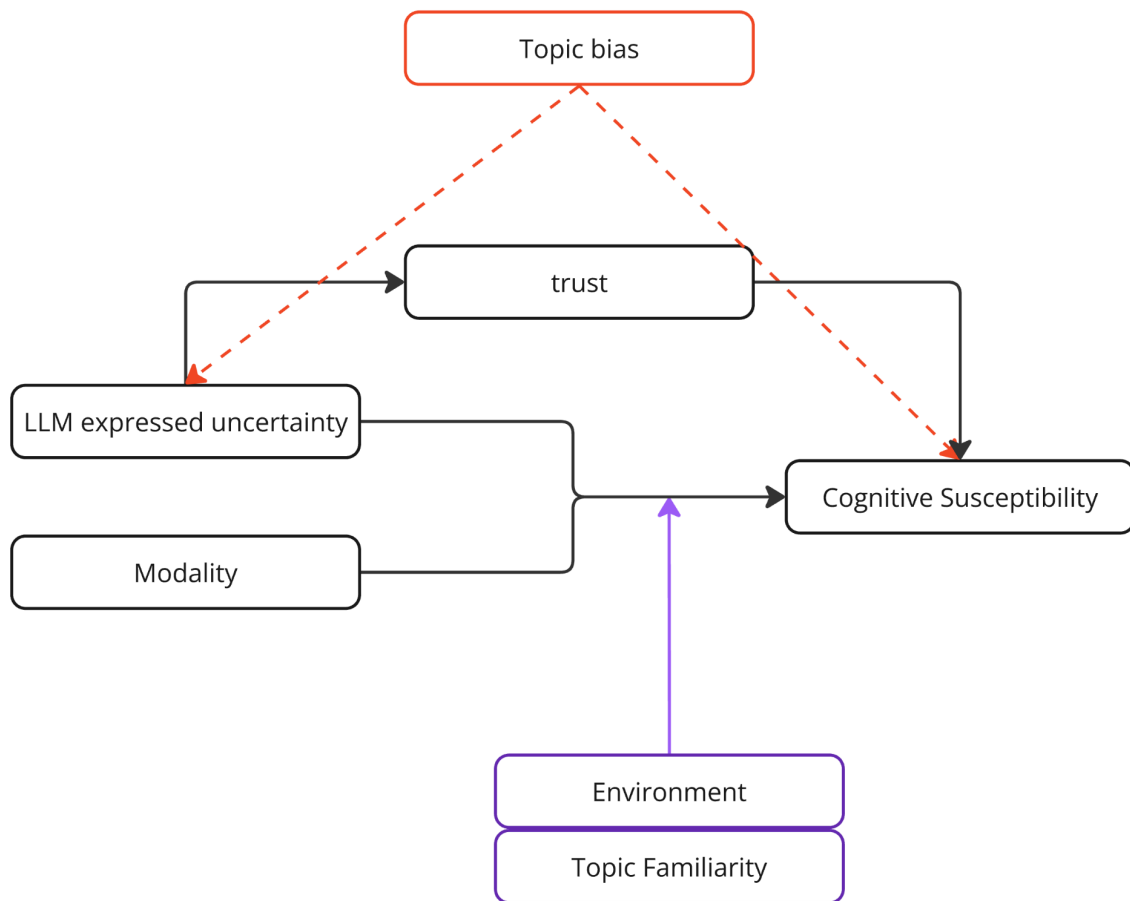
Q: Go through the topic of today

Q: Do we want consonant/dissonant? Let us do neutral for now

✓ The result of this study aims to uncover how users cognitive susceptibility can change with voice based interactions with conversational search grounded in context realism. This study will also look at the landscape of potential solutions developed so far and their effectiveness in this modality and recommendations for future developments.

~~The result of this study will give us insights into the expressive and persuasive power of LLMs voice modality and what are the gaps we need to address to tackle the over-reliance problem in this modality. Finally, this study provides context realism by exploring the interactions with voice agents in the environments which they are used for.~~

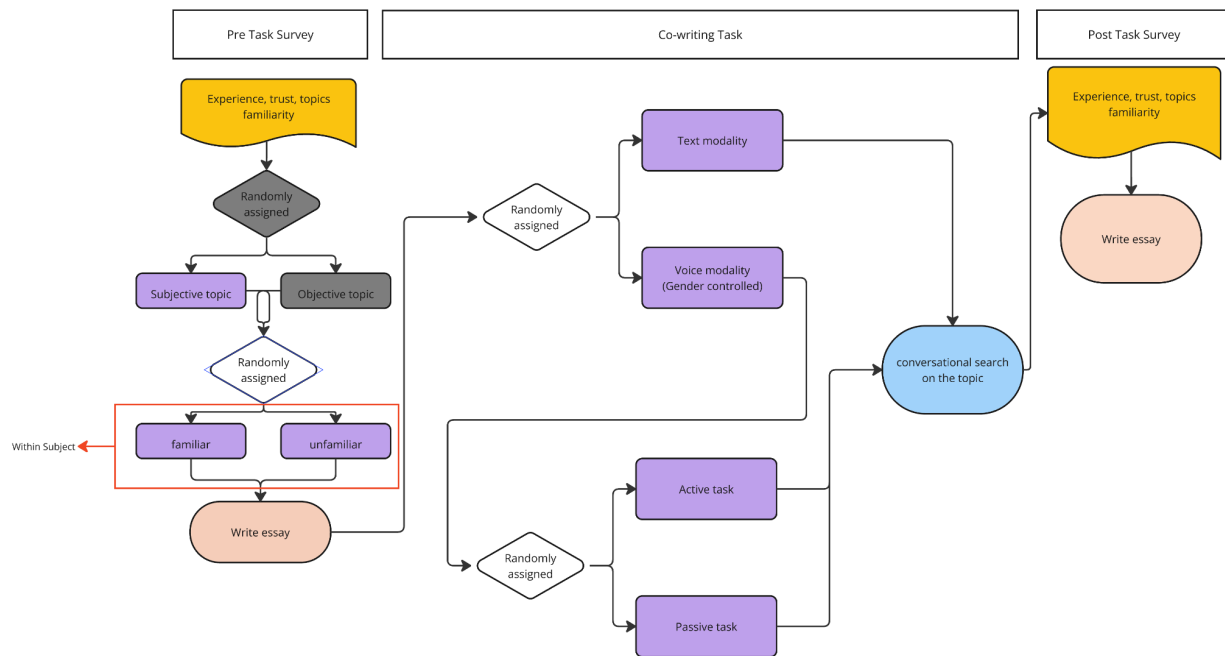
Variables



Process

Active Task: An active task is a task with no distractors i.e. the user will be solely focused on the task of information seeking with the conversational agent.

Passive Task: Passive task involves a distractor i.e. the user is focused on another task such as playing a game, driving, or exercising and simultaneously interacting with the voice agent for information seeking. The purpose of this task is to simulate a real world environment for voice-based agents where users are occupied with tasks because of which they cannot use their hands or look at the screen to interact with the agent.



Study design

flow : survey1, between subject (topic1), survey2, between subject bias of chatbot (consonant, dissonant or neutral), Within subject randomly assign (task1 or task2), survey 3, within subject (survey 4, survey 5)

Survey 1:

{

```

  "id": "demographics",
  "type": "survey",
  "title": "Demographic Survey",
  "questions": [
    {
      "id": "gender",
      "type": "singleChoice",
      "prompt": "What's your gender?",
      "options": [
        "Female",

```

```
"Male",
"Non-binary/third gender",
"Prefer not to say"
]
},
{
  "id": "ageGroup",
  "type": "singleChoice",
  "prompt": "What's your age?",
  "options": [
    "Under 18",
    "18-24",
    "25-34",
    "35-44",
    "45-54",
    "55-64",
    "65-74",
    "75-84",
    "85 or older",
    "Prefer not to say"
  ]
},
{
  "id": "education",
  "type": "singleChoice",
  "prompt": "What's your level of education?",
  "options": [
    "Less than high school diploma",
    "High school diploma or equivalent",
    "Bachelor's degree (e.g. BA or BS)",
    "Master's degree (e.g. MA, MS or MEd)",
    "Doctorate (e.g. PhD, EdD)",
    "Other",
    "Prefer not to answer"
  ]
}
```

```

    }
  ]
},

{
  "id": "cai_semantic",
  "type": "survey",
  "title": "Attitudes toward Conversational AI",
  "scale": 7,
  "questions": [
    { "id": "sd_useful", "type": "bipolar", "left": "Useful", "right": "Useless" },
    { "id": "sd_pleasant", "type": "bipolar", "left": "Pleasant", "right": "Unpleasant" },
    { "id": "sd_effective", "type": "bipolar", "left": "Effective", "right": "Ineffective" },
    { "id": "sd_bad", "type": "bipolar", "left": "Bad", "right": "Good" },
    { "id": "sd_nice", "type": "bipolar", "left": "Nice", "right": "Annoying" },
    { "id": "sd_irritating", "type": "bipolar", "left": "Irritating", "right": "Likeable" },
    { "id": "sd_helpful", "type": "bipolar", "left": "Helpful", "right": "Worthless" },
    { "id": "sd_undesirable", "type": "bipolar", "left": "Undesirable", "right": "Desirable" },
    { "id": "sd_stimulating", "type": "bipolar", "left": "Stimulating", "right":
"Sleep-inducing" },
    { "id": "sd_trustworthy", "type": "bipolar", "left": "Trustworthy", "right": "Unreliable" }
  ]
},

{
  "id": "cai_use_frequency",
  "type": "survey",
  "title": "Conversational-AI Usage",
  "questions": [
    {
      "id": "cai_freq",
      "type": "singleChoice",
      "prompt": "How often did you use conversational AIs for information needs?",
      "options": [
        "Never",

```



```
"At least once per month",
"At least once per week",
"At least once per day",
"Multiple times per day"
]
}
]
},
```

Between subject randomly assign topic1 [678bae020321fca7685c8cab,
678c6e940321fca7685c8cac, 6480a075ae6e9d404d974c01] These are moongoDB ids

Survey2:

Instructions: Please answer the questions based on the topic you have been assigned.

```
"dynamicContent": "{{topicDescription}}"
"questions": [

{

  "id": "familiarity_pre",

  "type": "singleChoice",

  "prompt": "How familiar are you with the topic?",

  "options": [

    "Very unfamiliar",

    "Unfamiliar",

    "Somewhat familiar",

    "Familiar",

    "Very familiar"
```

```
]
},
{
  "id": "stance_pre",
  "type": "singleChoice",
  "prompt": "What is your stance on the topic?",
  "options": [
    "Strongly Disagree",
    "Disagree",
    "Somewhat Disagree",
    "Somewhat Agree",
    "Agree",
    "Strongly Agree"
  ]
}
```

Write a short essay about what you know about the topic and your stance on the topic.

Answer: text area

Minimum: 500 words

Task1:

_id: 1232432423

In this task1 the user will interact with the chatbot with rag setting with the bias of the bot assigned. The only features will be rag and the input will be simple input without any tool calling options. The layout should be simple chat layout (no additional scaffolds). Number of required turns: four. Filter for checking the question is a valid

question: yes. Stream: yes.

Backend route prefix: /12312312 (there should be a list of supported study prefixes, for instance this might be for simple-rag-chat - we move that to a folder inside the handler with the same name so it will be /openai/controller/handlers/12312312/[chat.ts](#) where we will have the [simple-rag-chat.ts](#)

frontend route pr: /12312312

This will be similar to the backend route where we will have a folder and route with /study/12312312/[userId]/page.ts

Task 2:

_id: 3rdew24

In this task2 the user will interact with voice chat in a streaming mode. We can say the interface should look like a voice bubble talking and a way to start and stop the voice chat. It is similar to what we have in /voice-chat/[id]

Survey 3:

Instructions: Please answer the questions based on the topic you have been assigned.

```
"dynamicContent": "{{topicDescription}}"
```

```
"questions": [
```

```
{
```

```
  "id": "familiarity_pre",
```

```
  "type": "singleChoice",
```

```
  "prompt": "How familiar are you with the topic?",
```

```
  "options": [
```

```
    "Very unfamiliar",
```

```
    "Unfamiliar",
```

```
    "Somewhat familiar",
```

```
    "Familiar",
```

```
"Very familiar"

]

},

{

  "id": "stance_pre",

  "type": "singleChoice",

  "prompt": "What is your stance on the topic?",

  "options": [

    "Strongly Disagree",

    "Disagree",

    "Somewhat Disagree",

    "Somewhat Agree",

    "Agree",

    "Strongly Agree"

  ]

}
```

Write a short essay about what you know about the topic and your stance on the topic.

Answer: text area

Minimum: 500 words

Survey 4:

Instructions: rate the article

Article: U.S. health care spending rose 2.7% in 2021 to a total of \$4.3 trillion nationally and accounted for 18.3% of the U.S. Gross Domestic Product (GDP). Of the \$4.3 trillion, health insurance paid 71% (with private insurance paying 28%, and government

programs covering the rest via Medicare, Medicaid, and other programs), patients paid 10% as did other third party payers (such as the Indian Health Service), investment covered 5%, and government public health activities accounted for 4%. Per person, health care costs averaged \$12,914.

```
"title": "Dissonant Article Rating",
```

```
"questions": [
```

```
{
```

```
  "id": "position_extreme",
```

```
  "type": "singleChoice",
```

```
  "prompt": "The position reflected in the article is extreme.",
```

```
  "options": [
```

```
    "Strongly Disagree",
```

```
    "Disagree",
```

```
    "Neither Agree or Disagree",
```

```
    "Agree",
```

```
    "Strongly Agree"
```

```
  ]
```

```
},
```

```
{
```

```
  "id": "trust_article",
```

```
  "type": "singleChoice",
```

```
  "prompt": "I trust the article.",
```

```
"options": [  
  "Strongly Disagree",  
  "Disagree",  
  "Neither Agree or Disagree",  
  "Agree",  
  "Strongly Agree"  
]  
,  
{  
  "id": "agree_article",  
  "type": "singleChoice",  
  "prompt": "I agree with the article.",  
  "options": [  
    "Strongly Disagree",  
    "Disagree",  
    "Neither Agree or Disagree",  
    "Agree",  
    "Strongly Agree"  
  ]  
},  
{
```

```
"id": "tool_biased_against",  
  
"type": "singleChoice",  
  
"prompt": "I found the intelligent tool is biased against my stance.",  
  
"options": [  
  
    "Strongly Disagree",  
  
    "Disagree",  
  
    "Neither Agree or Disagree",  
  
    "Agree",  
  
    "Strongly Agree"  
  
    ],  
  
},  
  
{  
  
    "id": "tool_biased_toward",  
  
    "type": "singleChoice",  
  
    "prompt": "I found the intelligent tool is biased toward my stance.",  
  
    "options": [  
  
        "Strongly Disagree",  
  
        "Disagree",  
  
        "Neither Agree or Disagree",  
  
        "Agree",  
  
        "Strongly Agree"
```

```
]
}
]
```

Survey 5:

Instructions: rate the article

Article: Universal health care is an umbrella term for “a system that provides medical services to all people. The government offers it to everyone regardless of their ability to pay, and largely funds it through taxes,” according to economic expert Kimberly Amadeo.

```
"title": "Consonant Article Rating",
"questions": [
  {
    "id": "position_extreme",
    "type": "singleChoice",
    "prompt": "The position reflected in the article is extreme.",
    "options": [
      "Strongly Disagree",
      "Disagree",
      "Neither Agree or Disagree",
      "Agree",
```



```
    "Strongly Agree"
  ]
},
{
  "id": "trust_article",
  "type": "singleChoice",
  "prompt": "I trust the article.",
  "options": [
    "Strongly Disagree",
    "Disagree",
    "Neither Agree or Disagree",
    "Agree",
    "Strongly Agree"
  ]
},
{
  "id": "agree_article",
  "type": "singleChoice",
  "prompt": "I agree with the article.",
  "options": [
    "Strongly Disagree",
```

```
"Disagree",

"Neither Agree or Disagree",

"Agree",

"Strongly Agree"

]

},

{

  "id": "tool_biased_against",

  "type": "singleChoice",

  "prompt": "I found the intelligent tool is biased against my stance.",

  "options": [

    "Strongly Disagree",

    "Disagree",

    "Neither Agree or Disagree",

    "Agree",

    "Strongly Agree"

  ]

},

{

  "id": "tool_biased_toward",

  "type": "singleChoice",
```

```
"prompt": "I found the intelligent tool is biased toward my stance.",  
"options": [  
    "Strongly Disagree",  
    "Disagree",  
    "Neither Agree or Disagree",  
    "Agree",  
    "Strongly Agree"  
]  
}  
]
```

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research on scarcity, cognitive load reliance on ai

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Pattern is a vague word
Reliance is a decision making task
Make it more concrete

Voice
Active vs passive how to coll

Cognitive constraints

Within subject – one familiar and one unfamiliar